

Clustering and Hierarchical Clustering Questions and Answers

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1. What is clustering in machine learning?

- Clustering is an unsupervised learning technique used to group similar data points into clusters based on their features. It aims to identify natural groupings within data without prior labeling.

2. How does clustering differ from classification?

- Clustering is unsupervised, meaning it identifies groups without predefined labels, while classification is supervised and uses labeled data to predict the category of new data points.

3. What are some common clustering algorithms?

- Popular clustering algorithms include K-means, hierarchical clustering, DBSCAN, and Gaussian mixture models (GMMs).

4. What is the K-means clustering algorithm?

- K-means is a partition-based clustering algorithm that assigns data points to the nearest cluster center (centroid) and iteratively updates centroids to minimize within-cluster variance.

5. What are the main steps in the K-means algorithm?

- The K-means algorithm follows these steps:

1. Choose the number of clusters (K).
2. Initialize centroids randomly.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids based on the mean of assigned points.
5. Repeat steps 3 and 4 until centroids stabilize.

6. What is the objective function of K-means clustering?

- The objective function of K-means is to minimize the sum of squared distances between each data point and its corresponding cluster centroid.

7. What is the difference between K-means and K-medoids clustering?

- K-means uses the mean of points as centroids, while K-medoids uses actual data points as cluster centers, making it more robust to noise and outliers.

8. What is hierarchical clustering?

- Hierarchical clustering is a clustering approach that builds a hierarchy of clusters using either a bottom-up (agglomerative) or top-down (divisive) approach.

9. What are the types of hierarchical clustering?

- The two main types are:
 - Agglomerative (bottom-up): Each data point starts as its own cluster, and clusters are merged iteratively.
 - Divisive (top-down): All data points start in one cluster, which is repeatedly split until only single points remain.

10. What is a dendrogram?

- A dendrogram is a tree-like diagram that shows the hierarchical relationships between clusters and can be used to visualize the clustering process.